

DAY: _____

DATE: _____

Name: Muhammad Muzammil

Section: A

Program: Applied Computing

Assignment 01:

Ride Sharing System

A: Purpose:

A ride sharing system is an web platform that connects passengers who need a ride, with nearby devices through an app.

A ride sharing system handles ride requests, driver matching, live navigation, fare (connection) calculation, payments and ratings.

The goal of a ride sharing system is to make transport convenient, affordable and safe while giving a drivers way to earn.

B: Users:

Passengers want to request rides they want to track the drivers they want to pay for the trip

Drivers need to accept request they need to navigate to the destination and they need to receive earnings

Administrators and support staff have to manage accounts they have to resolve complaints. They have to monitor operations

C: Stakeholders:

- Passengers and drivers
- The ride sharing company (owners, management, investors)
- Development, operations and customer support teams
- Payment providers and banks
- Government and transport regulators

- Map and navigation service providers
- Local communities and traffic authorities

D: 5 functional Requirements:

1. The system shall allow users to register and log in as a passenger or driver, with identity and vehicle verification for drivers.
2. The system shall allow passengers to request a ride by entering pickup and drop-off location and see an estimated fare.
3. The system shall automatically match a ride request with the nearest driver available, who can accept it or reject it.
4. The system shall provide real-time GPS tracking of ride for both passenger and driver.

5. The system process payment, generate a receipt, and allow both sides of rate each other

E: 5 Non-Functional requirement:

1. **Performance:** Driver matching should complete within second.
2. **Security:** payment and personal data must be encrypted, and the system should follow secure authentication.
3. **Availability:** The system should be available 24/7 with at least 99.9% uptime.
4. **Scalability:** It should handle thousands of simultaneous ride requests during peak hours, rain or event.
5. **Usability:** The app should be enough to book a ride in a few taps and support multiple languages including urdu.

F: Three Risks:

1. **Safety Risks:** passengers or driver may face accidents or frauds, especially during night rides.

2. **Data privacy and security breach:**

Location history, phone numbers, and payment details are highly sensitive and could be leaked.

3. **System failure and GPS inaccuracy:**

Server downtime, poor internet or wrong GPS data can cause fake bookings, wrong pickups or incorrect fares.

G: Three quality attribute:

1. **Reliability:** The system consistently matches rides and complete trips without failure and lost data.

2. Security and safety:

User data, payments and physical safety are protected

3. Performance efficiency:

Fast matching, accurate ETAs, and smooth real time tracking

H.: Where AI could be used:

1. Smart driver matching:

Picking the best drivers by distance, traffic and rating, not just proximity.

2. Dynamic Pricing:

Adjusting fares based on demand and supply.

3. ETA and route prediction:

Predicting travel time and avoiding congestion using traffic data.

4. Fraud detection: spotting fake accounts, fake trips and payments fraud.

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I What Could Go Wrong if
AI is used incorrectly

- 1) Unfair pricing
- 2) Biased matching or deactivation
- 3) Privacy violation
- 4) Wrong route
- 5) False fraud flag
- 6) Missed safety threats
- 7) Lack of transparency.